KUKU AI GENERALIST ASSIGNMENT - SOLUTION DOCUMENT

Problem 2: DepositMitra (AI Paralegal for Indian Renter Deposit Recovery)

1. Problem Choice and Framing

Why This Problem?

In urban India (Bengaluru, Pune, Gurgaon, Mumbai), security deposit retention is an endemic crisis. Over 80% of young renters forfeit Rs. 25,000 to 80,000 upon moving out due to arbitrary deductions: routine repainting charges, unjustified cleaning fees, or outright landlord ghosting. Formal legal recourse is economically broken: retaining an advocate costs Rs. 15,000-30,000 upfront for a Rs. 40,000 dispute. Intimidated by opaque jargon and court backlogs, ordinary citizens abandon their claims.

Scoping & Intentional Boundaries (Depth over Shallow Breadth):

Rather than building a superficial general lawyer covering murder, divorce, and corporate law, DepositMitra is laser-focused 100% on Residential Rental Security Deposit Disputes.

- In-Scope: Unlawful painting/cleaning deductions, ghosting post-key handover, notice period disputes, wear-and-tear classification, demand notice generation, and negotiation coaching.
- Strict Out-of-Scope (Safety Triage): Physical threats, goons, forced lockouts, and commercial leases (>Rs. 50L). These trigger immediate deterministic hard-refusal and direct referral to emergency helplines (Dial 112) or the District Legal Services Authority (DLSA).

Assumptions Made:

(1) Ordinary renters communicate in colloquial Hinglish or informal text; (2) Landlords count on tenant legal hesitation; (3) Painting and minor weathering constitute 'Normal Wear & Tear' under Indian law unless the landlord substantiates deliberate damage with move-in inventories and third-party GST invoices.

2. Approach and Scale: What We Gave to AI vs. Deterministic Rules

To eliminate hallucination of legal sections and fabricated court cases, DepositMitra balances deterministic code with LLM reasoning:

What AI Does (Gemini 1.5/2.5 Flash):	What Deterministic Code / Rules Do:
- Parses messy Hinglish & conversational stories	- Safety triage: Regex catches criminal threats
- Extracts chronology & detects missing proof gaps	- Statutory Registry: Whitelist of verified Indian Acts
- Plain-language legal advice & strategic standing	- Anti-Hallucination Filter: Blocks fake case citations
- Interactive stubborn landlord persona roleplay	- Limitation Audit: Enforces 3-year statutory limit
- Drafts tailored, persuasive 15-day notice clauses	- Unit economics: Zero-cost rule execution & fallback

Human in the Loop & Unit Economics at Scale:

- Human in the Loop: The tenant reviews and customizes the generated demand notice before dispatch. A clear statutory disclaimer clarifies that DepositMitra is a pre-litigation paralegal assistant, not an advocate.
- Real-World Economics: A full 5-phase consultation consumes ~3,800 tokens. On Gemini 1.5 Flash, the cost is approx. $0.0019 USD (< 16 paise INR) per case. Scaling to 100,000 cases costs under Rs. 16,000 (~$190), making it economically deployable as a free public utility.

3. Quality Checking, Real Runs & Honest Failure Modes

The system was evaluated against 5 realistic test cases (unedited raw logs in /raw_outputs/):

Case ID	Scenario & City	Engine Action & Output	Status / Leverage
CASE-01	Bengaluru: Rs.35k deducted for painting without bills	Sanitized 15-day notice citing Sec 73 ICA & MTA	PASS (95/100 Leverage)
CASE-02	Pune: Landlord ghosting after handover (50 days)	Demanded 18% statutory interest & CPC summary suit	PASS (95/100 Leverage)
CASE-03	Gurgaon: Notice dispute (written 1 mo vs oral 2 mo)	Voided arbitrary penalty under Sec 74 ICA	PASS (75/100 Leverage)
CASE-04	Mumbai: Seepage & 8-yr geyser wear charged to tenant	Applied Normal Wear & Tear doctrine (Sec 15 MTA)	PASS (95/100 Leverage)
CASE-05	Noida: Landlord sent goons, threatened violence	Hard stop: Civil processing halted; Dial 112 triage	PASS (Emergency Alert)

Where the Tool Fails Honestly (Boundary Limitations):

1. Cash-Only Oral Leases: Without banking records or a signed lease, the tool cannot manufacture documentary proof. It alerts the tenant to banking gaps but cannot provide high legal leverage.
2. Co-Mingled Genuine Damage: If a tenant genuinely broke a fixture but the landlord inflated the repair cost by 5x, the tool cannot verify physical damage without on-site computer vision or contractor quote audits.
3. Local Rent Control Nuances: Certain states (e.g. Maharashtra Rent Control Act vs Karnataka Rent Act) have distinct local forum procedural requirements. The tool recommends central summary suits and DLSA mediation, flagging when regional court filings require local bar counsel.

4. The Prototype & Reproducibility

- Structure: Clean Python architecture separating statutes, guardrails, intake, evaluation, notice generation, and negotiation simulation.
- Model Swapping & Zero-Cost Run: Fully operational in 100% offline heuristic mode without API keys, while seamlessly connecting to Google Gemini (or interview APIs) when provided.
- Quick Run Instructions:

```
$ cd KUku && pip install -r requirements.txt
$ python3 scripts/run_eval.py    # Runs batch evaluation & regenerates raw outputs
$ streamlit run app.py           # Launches interactive web application
```

5. Conclusion & Product Vision

DepositMitra bridges India's pre-litigation justice gap by delivering credible, anti-hallucinatory legal leverage. By arming tenants with statutory facts, 15-day demand notices, and roleplay confidence, it enables ordinary citizens to recover their hard-earned money without entering protracted courtroom battles.